

Fraction Universe

Every graded set, every week — 6 weeks, 30 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

WEEK 6

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

One Whole, Cut Up

Answers are whole numbers — when a question asks for a numerator, type just that number.

Warm-Up 12 ITEMS • RECALL

1 Halves in one whole

2 Quarters in one whole

3 Eighths in one whole

4 $1/2 = ?/4 \rightarrow$ numerator

5 $1/2 = ?/8 \rightarrow$ numerator

6 $3/3 =$ how many wholes?

7 Thirds in one whole

8 Fifths in one whole

9 Sixths in one whole

10 $1/3 = ?/6 -$ numerator

11 $1/5 = ?/10 -$ numerator

12 Tenths in one whole

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $1/4 + 1/4 + 1/4 = ?/4 \rightarrow$ numerator (Count the quarters)

.....

14. How many thirds in 2 wholes? (Three in each whole)

.....

15. $5/5 =$ how many wholes? (All five pieces)

.....

16. $7/4 -$ how many whole ones? ($4/4$ makes one whole)

.....

17. $7/4 -$ how many quarters left over? ($7 - 4$)

.....

18. $3/4 = ?/12 -$ numerator

.....

19. $2/5 = ?/20 -$ numerator

.....

20. $5/6 = ?/12 -$ numerator

.....

21. $7/8 = ?/16 -$ numerator

.....

22. $2/3 = ?/15 -$ numerator

.....



Working space



Error journal

Same Number, New Name

Multiply top and bottom by the same thing and nothing about the amount changes.

Warm-Up 12 ITEMS • RECALL

1 $1/2 = ?/6 \rightarrow$ numerator

2 $1/3 = ?/6 \rightarrow$ numerator

3 $2/3 = ?/6 \rightarrow$ numerator

4 $1/4 = ?/8 \rightarrow$ numerator

5 $3/4 = ?/8 \rightarrow$ numerator

6 $1/5 = ?/10 \rightarrow$ numerator

7 $1/2 = ?/10 \rightarrow$ numerator

8 $1/4 = ?/12 \rightarrow$ numerator

9 $1/3 = ?/9 \rightarrow$ numerator

10 $2/3 = ?/9 \rightarrow$ numerator

11 $3/5 = ?/10 \rightarrow$ numerator

12 $1/6 = ?/12 \rightarrow$ numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $3/4 = ?/12 \rightarrow$ numerator (The bottom was tripled)

14. $2/5 = ?/20 \rightarrow$ numerator (The bottom was multiplied by 4)

15. $6/8$ in quarters = $?/4 \rightarrow$ numerator (Divide both by 2)

16. $10/15$ simplified $\rightarrow$ numerator (Divide both by 5)

17. $9/12$ simplified $\rightarrow$ denominator (Divide both by 3)

18. $6/8 = ?/4 \rightarrow$ numerator

19. $10/15 = ?/3 \rightarrow$ numerator

20. $12/16 = ?/4 \rightarrow$ numerator

21. $9/15 = ?/5 \rightarrow$ numerator

22. $14/21 = ?/3 \rightarrow$ numerator



Which Is Bigger

Make the pieces the same size first. Then it's just counting.

Warm-Up 12 ITEMS • RECALL

1 Bigger: $\frac{1}{2}$ or $\frac{1}{4}$ → its denominator

2 Bigger: $\frac{1}{3}$ or $\frac{1}{5}$ → its denominator

3 Bigger: $\frac{3}{4}$ or $\frac{1}{4}$ → its numerator

4 Bigger: $\frac{2}{5}$ or $\frac{3}{5}$ → its numerator

5 Bigger: $\frac{5}{6}$ or $\frac{5}{8}$ → its denominator

6 Bigger: $\frac{7}{10}$ or $\frac{3}{10}$ → its numerator

7 Bigger: $\frac{1}{2}$ or $\frac{1}{5}$ — its denominator

8 Bigger: $\frac{1}{4}$ or $\frac{1}{6}$ — its denominator

9 Bigger: $\frac{4}{5}$ or $\frac{2}{5}$ — its numerator

10 Bigger: $\frac{5}{8}$ or $\frac{3}{8}$ — its numerator

11 Bigger: $\frac{1}{3}$ or $\frac{1}{10}$ — its denominator

12 Bigger: $\frac{7}{10}$ or $\frac{3}{10}$ — its numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{2}{3}$ or $\frac{3}{5}$ in fifteenths → bigger numerator (10/15 against 9/15)

14. $\frac{3}{4}$ or $\frac{5}{8}$ in eighths → bigger numerator (6/8 against 5/8)

15. $\frac{5}{6}$ or $\frac{7}{9}$ in eighteenths → bigger numerator (15/18 against 14/18)

16. Closer to 1: $\frac{7}{8}$ or $\frac{5}{6}$ → its denominator (A gap of $\frac{1}{8}$ beats a gap of $\frac{1}{6}$)

17. $\frac{2}{3}$ and $\frac{3}{4}$ in twelfths → bigger numerator (8/12 against 9/12)

18. Bigger: $\frac{2}{3}$ or $\frac{3}{5}$ in fifteenths — its numerator

19. Bigger: $\frac{3}{4}$ or $\frac{5}{8}$ in eighths — its numerator

20. Bigger: $\frac{5}{6}$ or $\frac{4}{5}$ in thirtieths — its numerator

21. Bigger: $\frac{1}{2}$ or $\frac{4}{9}$ in eighteenths — its numerator

22. Bigger: $\frac{7}{12}$ or $\frac{1}{2}$ in twelfths — its numerator

★ Challenge NOT GRADED • REASON IT OUT

23. Largest of $\frac{3}{5}$, $\frac{2}{3}$, $\frac{7}{10}$ in thirtieths → its numerator (18/30, 20/30, 21/30)

.....

24. Halfway between $\frac{1}{2}$ and $\frac{3}{4}$ as $\frac{?}{8}$ → numerator (Between $\frac{4}{8}$ and $\frac{6}{8}$)

.....

25. Largest of $\frac{5}{8}$, $\frac{2}{3}$, $\frac{7}{12}$ in twenty-fourths — its numerator

.....

26. Halfway between $\frac{1}{4}$ and $\frac{1}{2}$ as $\frac{?}{8}$ — numerator

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Adding Needs Matching Pieces

Change one fraction, or both, until the bottoms agree. Then count.

Warm-Up 12 ITEMS • RECALL

1 $1/4 + 1/4 = ?/4 \rightarrow$
numerator

2 $2/5 + 1/5 = ?/5 \rightarrow$
numerator

3 $3/8 + 2/8 = ?/8 \rightarrow$
numerator

4 $5/6 - 2/6 = ?/6 \rightarrow$
numerator

5 $7/10 - 4/10 = ?/10 \rightarrow$
numerator

6 $3/4 + 1/4 =$ how many
wholes?

7 $1/3 + 1/3 = ?/3 -$
numerator

8 $3/8 + 1/8 = ?/8 -$
numerator

9 $4/9 + 2/9 = ?/9 -$
numerator

10 $7/10 - 3/10 = ?/10 -$
numerator

11 $5/7 - 2/7 = ?/7 -$
numerator

12 $2/6 + 3/6 = ?/6 -$
numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $1/2 + 1/4 = ?/4 \rightarrow$ numerator ($2/4 + 1/4$)

14. $2/3 + 1/6 = ?/6 \rightarrow$ numerator ($4/6 + 1/6$)

15. $3/4 - 1/8 = ?/8 \rightarrow$ numerator ($6/8 - 1/8$)

16. $5/6 - 1/3 = ?/6 \rightarrow$ numerator ($5/6 - 2/6$)

17. $1/2 + 1/3 = ?/6 \rightarrow$ numerator ($3/6 + 2/6$)

18. $1/2 + 1/6 = ?/6 -$ numerator

19. $1/3 + 1/4 = ?/12 -$ numerator

20. $3/4 - 1/2 = ?/4 -$ numerator

21. $5/6 - 1/3 = ?/6 -$ numerator

22. $2/5 + 3/10 = ?/10 -$ numerator

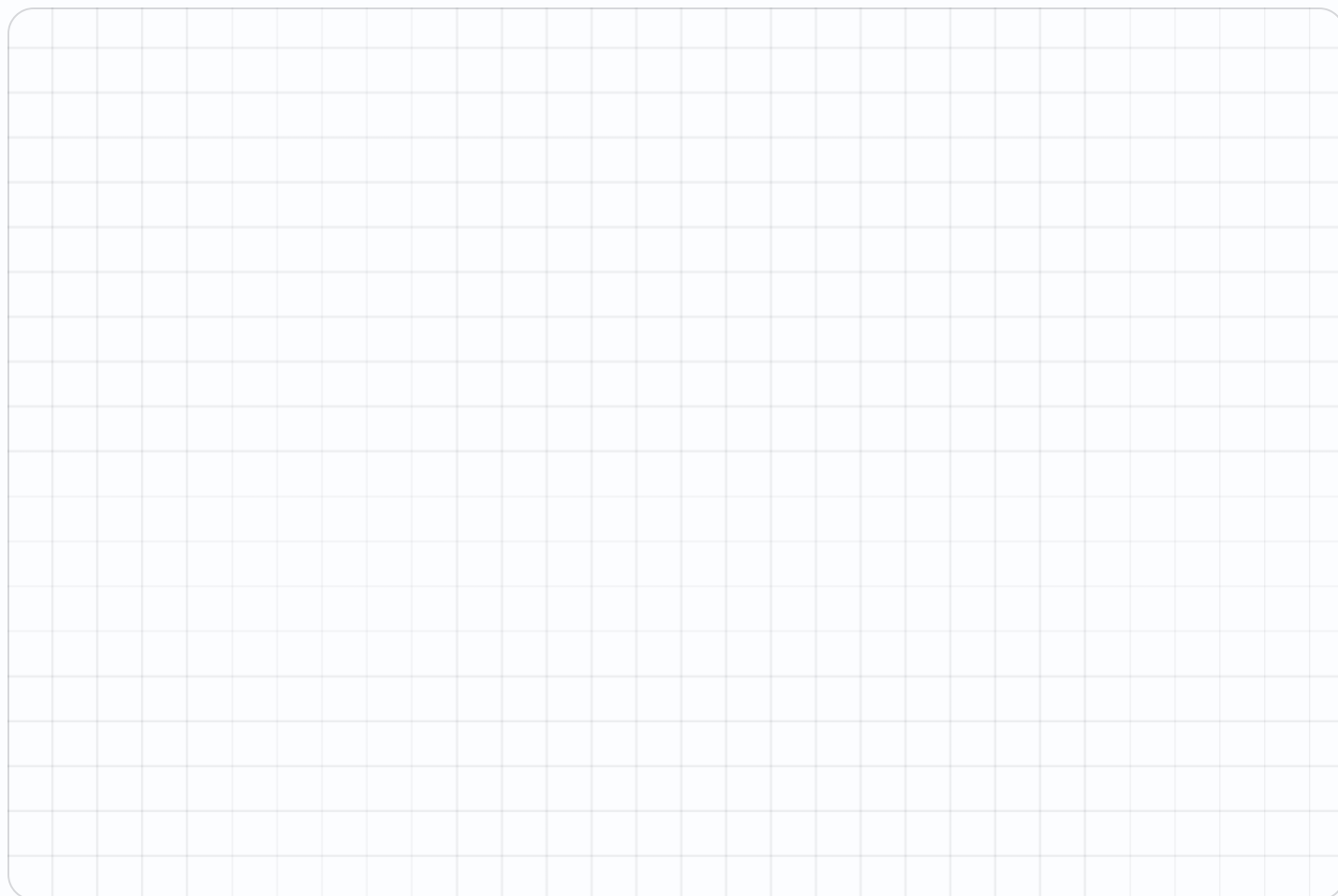
★ Challenge NOT GRADED • REASON IT OUT

23. $3/4 + 2/3 = ?/12 \rightarrow$ numerator (9/12 + 8/12)

24. $2 - 3/5 = ?/5 \rightarrow$ numerator (10/5 - 3/5)

25. $5/6 + 3/4 = ?/12$ — numerator

26. $3 - 2/5 = ?/5$ — numerator

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Fraction of a Set & Closest to One

Enrichment day. Divide by the bottom, multiply by the top — always that order.

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • One Whole, Cut Up

OUT OF 22

Warm-Up • 1-12 • 2 | 4 | 8 | 2 | 4 | 1 | 3 | 5 | 6 | 2 | 2 | 10

Core • 13-22 • 3 | 6 | 1 | 1 | 3 | 9 | 8 | 10 | 14 | 10

Challenge • 23-26 • 4 | 2 | 7 | 5

1.2 • Same Number, New Name

OUT OF 22

Warm-Up • 1-12 • 3 | 2 | 4 | 2 | 6 | 2 | 5 | 3 | 3 | 6 | 6 | 2

Core • 13-22 • 9 | 8 | 3 | 2 | 4 | 3 | 2 | 3 | 3 | 2

Challenge • 23-26 • 2 | 9 | 6 | 5

1.3 • Which Is Bigger

OUT OF 22

Warm-Up • 1-12 • 2 | 3 | 3 | 3 | 6 | 7 | 2 | 4 | 4 | 5 | 3 | 7

Core • 13-22 • 10 | 6 | 15 | 8 | 9 | 10 | 6 | 25 | 9 | 7

Challenge • 23-26 • 21 | 5 | 16 | 3

1.4 • Adding Needs Matching Pieces

OUT OF 22

Warm-Up • 1-12 • 2 | 3 | 5 | 3 | 3 | 1 | 2 | 4 | 6 | 4 | 3 | 5

Core • 13-22 • 3 | 5 | 5 | 3 | 5 | 4 | 7 | 1 | 3 | 7

Challenge • 23-26 • 17 | 7 | 19 | 13

Fri • Fraction of a Set & Closest to One

ENRICHMENT

Challenge • 1-14 • 5 | 12 | 15 | 30 | 35 | 24 | 10 | 19 | 10 | 8 | 14 | 30 | 18
| 24

Same Number, New Clothes

Multiply top and bottom by the same thing and the value never moves. Type fractions like $\frac{3}{4}$.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2} = \frac{?}{4}$ — the numerator

2 $\frac{1}{2} = \frac{?}{6}$

3 $\frac{1}{3} = \frac{?}{6}$

4 $\frac{1}{4} = \frac{?}{8}$

5 $\frac{2}{3} = \frac{?}{6}$

6 $\frac{3}{4} = \frac{?}{8}$

7 $\frac{1}{2} = \frac{?}{8}$

8 $\frac{1}{3} = \frac{?}{12}$

9 $\frac{1}{5} = \frac{?}{15}$

10 $\frac{2}{3} = \frac{?}{12}$

11 $\frac{3}{5} = \frac{?}{15}$

12 $\frac{1}{4} = \frac{?}{16}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{2}{5} = \frac{?}{15}$

14. $\frac{3}{8} = \frac{?}{24}$

15. $\frac{5}{6} = \frac{?}{18}$

16. $\frac{4}{7} = \frac{?}{21}$

17. $\frac{7}{10} = \frac{?}{100}$

18. $\frac{4}{5} = \frac{?}{25}$

19. $\frac{5}{8} = \frac{?}{24}$

20. $\frac{3}{7} = \frac{?}{21}$

21. $\frac{5}{9} = \frac{?}{27}$

★ Challenge NOT GRADED • REASON IT OUT

22. $9/12$ in simplest form — type as a/b

23. What was $1/2$ multiplied by to become $7/14$ — type the number

24. $15/20$ in simplest form — type as a/b

25. What was $1/3$ multiplied by to become $8/24$

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Simplest Form

Divide top and bottom by the same thing until nothing else fits.

Warm-Up 11 ITEMS • RECALL

1 $\frac{2}{4}$ in simplest form —
type as a/b

2 $\frac{3}{6}$

3 $\frac{4}{8}$

4 $\frac{2}{6}$

5 $\frac{5}{10}$

6 $\frac{3}{9}$

7 $\frac{6}{12}$ in simplest form —
type as a/b

8 $\frac{4}{6}$

9 $\frac{5}{15}$

10 $\frac{8}{10}$

11 $\frac{9}{12}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. $\frac{6}{8}$

.....

13. $\frac{10}{15}$

.....

14. $\frac{12}{16}$

.....

15. $\frac{18}{24}$

.....

16. $\frac{20}{25}$

.....

17. $\frac{21}{28}$

.....

18. $\frac{30}{45}$

.....

19. $\frac{14}{35}$

.....

20. $\frac{27}{36}$

.....

Spot the Equivalent

Cross-multiply, or find a common denominator. Either proof counts.

Warm-Up 12 ITEMS • RECALL

1 Is $\frac{1}{2}$ the same as $\frac{2}{4}$ — yes or no

2 Is $\frac{1}{3}$ the same as $\frac{2}{6}$

3 Is $\frac{1}{2}$ the same as $\frac{1}{3}$

4 Is $\frac{3}{4}$ the same as $\frac{6}{8}$

5 Is $\frac{2}{5}$ the same as $\frac{4}{10}$

6 Is $\frac{2}{3}$ the same as $\frac{3}{4}$

7 Is $\frac{1}{4}$ the same as $\frac{2}{8}$ — yes or no

8 Is $\frac{2}{3}$ the same as $\frac{4}{6}$

9 Is $\frac{1}{2}$ the same as $\frac{2}{5}$

10 Is $\frac{3}{5}$ the same as $\frac{6}{10}$

11 Is $\frac{1}{3}$ the same as $\frac{3}{9}$

12 Is $\frac{2}{4}$ the same as $\frac{3}{5}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is $\frac{6}{9}$ the same as $\frac{8}{12}$ — yes or no

.....

14. Is $\frac{5}{8}$ the same as $\frac{10}{18}$

.....

15. Is $\frac{9}{12}$ the same as $\frac{15}{20}$

.....

16. $\frac{1}{2} \times \frac{5}{5}$ — type as a/b

.....

17. Is $\frac{7}{21}$ the same as $\frac{1}{3}$

.....

18. Is $\frac{6}{9}$ the same as $\frac{8}{12}$

.....

19. Is $\frac{5}{8}$ the same as $\frac{10}{16}$

.....

20. Is $\frac{4}{7}$ the same as $\frac{8}{15}$

.....

21. Is $\frac{9}{15}$ the same as $\frac{12}{20}$

.....

22. Is $\frac{7}{9}$ the same as $\frac{14}{17}$

.....

Fractions Equal to One

Any fraction with matching top and bottom is just 1 wearing a costume.

Warm-Up 12 ITEMS • RECALL

1 $5/5$ as a whole number2 $8/8$ 3 $12/12$ 4 $4/4$ 5 $100/100$ 6 $2/2$ 7 $7/7$ as a whole number8 $9/9$ 9 $6/6$ 10 $20/20$ 11 $3/3$ 12 $50/50$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $6/3$ as a whole number14. $12/4$ 15. $20/5$

16. How many quarters make 2

17. How many thirds make 3

18. $10/5$ as a whole number19. $18/6$ 20. $24/8$ 21. $9/2$ as a mixed number — the whole part22. $13/5$ — the whole part



Working space



Error journal

Equivalence Wall

Build the strips and find every pair that lines up.

Warm-Up 4 ITEMS • RECALL

1 Halves in a whole

2 Quarters in a whole

3 Sixths in a whole

4 Tenths in a whole

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Quarters that line up with $\frac{1}{2}$ — type as a/b

.....

6. Sixths that line up with $\frac{1}{3}$

.....

7. $\frac{1}{2} = ?/12$

.....

8. $\frac{2}{3} = ?/12$

.....

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Same Number, New Clothes

OUT OF 21

Warm-Up • 1-12 • 2 | 3 | 2 | 2 | 4 | 6 | 4 | 4 | 3 | 8 | 9 | 4

Core • 13-21 • 6 | 9 | 15 | 12 | 70 | 20 | 15 | 9 | 15

Challenge • 22-25 • $\frac{3}{4}$ | 7 | $\frac{3}{4}$ | 8

2.2 • Simplest Form

OUT OF 20

Warm-Up • 1-11 • $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{3}$ | $\frac{4}{5}$ | $\frac{3}{4}$

Core • 12-20 • $\frac{3}{4}$ | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{4}{5}$ | $\frac{3}{4}$ | $\frac{2}{3}$ | $\frac{2}{5}$ | $\frac{3}{4}$

Challenge • 21-24 • $\frac{3}{4}$ | 12 | $\frac{3}{4}$ | 15

2.3 • Spot the Equivalent

OUT OF 22

Warm-Up • 1-12 • yes | yes | no | yes | yes | no | yes | yes | no | yes | yes | no

Core • 13-22 • yes | no | yes | $\frac{5}{10}$ | yes | yes | yes | no | yes | no

Challenge • 23-26 • 9 | yes | 6 | yes

2.4 • Fractions Equal to One

OUT OF 22

Warm-Up • 1-12 • 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1

Core • 13-22 • 2 | 3 | 4 | 8 | 9 | 2 | 3 | 3 | 4 | 2

Challenge • 23-26 • 3 | 2 | 5 | 3

Fri • Equivalence Wall

OUT OF 8

Warm-Up • 1-4 • 2 | 4 | 6 | 10

Core • 5-8 • $\frac{2}{4}$ | $\frac{2}{6}$ | 6 | 8

Challenge • 9-14 • $\frac{9}{12}$ | $\frac{8}{12}$ | 12 | 12 | $\frac{2}{3}$ | 7

Common Denominators

Cut both into the same size pieces, then just compare the counts.

Warm-Up 11 ITEMS • RECALL

1 Common denominator of $\frac{1}{2}$ and $\frac{1}{4}$

2 Common denominator of $\frac{1}{3}$ and $\frac{1}{6}$

3 Common denominator of $\frac{1}{2}$ and $\frac{1}{3}$

4 $\frac{1}{2}$ as quarters — the numerator

5 $\frac{1}{3}$ as sixths — the numerator

6 Common denominator of $\frac{1}{4}$ and $\frac{1}{8}$

7 Common denominator of $\frac{1}{2}$ and $\frac{1}{6}$

8 Common denominator of $\frac{1}{3}$ and $\frac{1}{9}$

9 $\frac{1}{2}$ as sixths — the numerator

10 $\frac{1}{3}$ as ninths — the numerator

11 $\frac{1}{4}$ as eighths — the numerator

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Common denominator of $\frac{2}{3}$ and $\frac{3}{4}$

13. Common denominator of $\frac{3}{5}$ and $\frac{1}{2}$

14. $\frac{2}{3}$ as twelfths — the numerator

15. $\frac{3}{4}$ as twelfths — the numerator

16. Common denominator of $\frac{5}{6}$ and $\frac{3}{8}$

17. Smallest common denominator of $\frac{1}{4}$ and $\frac{1}{6}$

18. Of $\frac{1}{3}$ and $\frac{1}{5}$

19. Of $\frac{2}{5}$ and $\frac{3}{10}$

20. Of $\frac{3}{8}$ and $\frac{1}{6}$

21. Of $\frac{5}{12}$ and $\frac{1}{8}$

★ Challenge NOT GRADED • REASON IT OUT

22. Smallest common denominator of $\frac{1}{4}$, $\frac{1}{6}$ and $\frac{1}{8}$

23. Common denominator of $\frac{7}{10}$ and $\frac{2}{15}$

24. Smallest common denominator of $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{6}$

25. Common denominator of $\frac{5}{9}$ and $\frac{7}{12}$

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Benchmarks

Nearer 0, a half, or 1? Decide before you calculate anything.

Warm-Up 12 ITEMS • RECALL

1 Is $\frac{1}{8}$ nearer 0 or 1 — type 0 or 1

2 Is $\frac{7}{8}$ nearer 0 or 1

3 $\frac{4}{8}$ simplified — type as a/b

4 Is $\frac{5}{6}$ nearer $\frac{1}{2}$ or 1 — type $\frac{1}{2}$ or 1

5 Is $\frac{1}{5}$ nearer 0 or $\frac{1}{2}$ — type 0 or $\frac{1}{2}$

6 Is $\frac{3}{5}$ more or less than $\frac{1}{2}$ — type more or less

7 Is $\frac{1}{10}$ nearer 0 or 1 — type 0 or 1

8 Is $\frac{9}{10}$ nearer 0 or 1

9 $\frac{5}{10}$ simplified — type as a/b

10 Is $\frac{1}{6}$ nearer 0 or $\frac{1}{2}$ — type 0 or $\frac{1}{2}$

11 Is $\frac{3}{5}$ nearer $\frac{1}{2}$ or 1 — type $\frac{1}{2}$ or 1

12 $\frac{6}{12}$ simplified — type as a/b

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is $\frac{5}{9}$ more or less than $\frac{1}{2}$

.....

14. Is $\frac{4}{9}$ more or less than $\frac{1}{2}$

.....

15. Is $\frac{7}{16}$ more or less than $\frac{1}{2}$

.....

16. Is $\frac{9}{16}$ more or less than $\frac{1}{2}$

.....

17. Nearest benchmark to $\frac{11}{12}$ — type 0, $\frac{1}{2}$ or 1

.....

18. Is $\frac{11}{20}$ more or less than $\frac{1}{2}$

.....

19. Nearest benchmark to $\frac{9}{10}$ — type 0, $\frac{1}{2}$ or 1

.....



Working space



Error journal

Comparing & Ordering

Common denominator, or reason from benchmarks. Say which you used.

Warm-Up 12 ITEMS • RECALL

1 Larger: $\frac{1}{2}$ or $\frac{1}{3}$ — type it

2 Larger: $\frac{3}{4}$ or $\frac{1}{2}$

3 Larger: $\frac{1}{4}$ or $\frac{1}{8}$

4 Larger: $\frac{2}{3}$ or $\frac{1}{3}$

5 Smaller: $\frac{1}{5}$ or $\frac{1}{6}$

6 Larger: $\frac{5}{8}$ or $\frac{3}{8}$

7 Larger: $\frac{1}{2}$ or $\frac{1}{4}$ — type it

8 Larger: $\frac{2}{3}$ or $\frac{1}{2}$

9 Larger: $\frac{3}{5}$ or $\frac{2}{5}$

10 Larger: $\frac{1}{3}$ or $\frac{1}{4}$

11 Smaller: $\frac{1}{8}$ or $\frac{1}{10}$

12 Larger: $\frac{5}{6}$ or $\frac{1}{2}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Larger: $\frac{2}{3}$ or $\frac{3}{4}$ — type it

14. Larger: $\frac{3}{5}$ or $\frac{5}{8}$

15. Larger: $\frac{5}{6}$ or $\frac{7}{9}$

16. Smallest of $\frac{1}{3}$, $\frac{2}{5}$, $\frac{1}{4}$ — type it

17. Largest of $\frac{2}{3}$, $\frac{5}{8}$, $\frac{7}{12}$

18. Larger: $\frac{2}{3}$ or $\frac{5}{8}$ — type it

19. Larger: $\frac{3}{7}$ or $\frac{1}{2}$

20. Larger: $\frac{7}{12}$ or $\frac{5}{9}$

21. Larger: $\frac{4}{5}$ or $\frac{7}{9}$

22. Smaller: $\frac{5}{12}$ or $\frac{3}{8}$



Working space



Error journal

On the Number Line

Every fraction has one address between 0 and 1. Find it.

Warm-Up 10 ITEMS • RECALL

1 Halfway between 0 and 1
— type as a/b

2 Halfway between 0 and $\frac{1}{2}$

3 Halfway between $\frac{1}{2}$ and 1

4 Marks between 0 and 1 if
split into quarters

5 Marks if split into fifths

6 $\frac{1}{2}$ as a decimal

7 Halfway between 0 and $\frac{1}{4}$ — type as a/b

8 Halfway between $\frac{1}{4}$ and $\frac{1}{2}$

9 Halfway between $\frac{3}{4}$ and 1

10 Marks between 0 and 1
if split into fifths

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $\frac{3}{4}$ as a decimal

12. $\frac{1}{4}$ as a decimal

13. $\frac{1}{5}$ as a decimal

14. $\frac{3}{5}$ as a decimal

15. Halfway between $\frac{1}{4}$ and $\frac{1}{2}$ — type as a/b

16. $\frac{1}{8}$ as a decimal

17. $\frac{1}{10}$ as a decimal

 Challenge NOT GRADED • REASON IT OUT

18. $\frac{7}{8}$ as a decimal

19. A fraction exactly between $\frac{1}{3}$ and $\frac{2}{3}$ — type as a/b

20. $\frac{3}{8}$ as a decimal

21. A fraction exactly between $\frac{1}{4}$ and $\frac{3}{4}$ — type as a/b

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 $1/2 = ?/8$ — the numerator

2 $4/8$ simplified — type as a/b

3 $1/3 = ?/9$ — the numerator

4 $6/8$ simplified — type as a/b

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $6/8$ simplified

6. Common denominator of $2/3$ and $3/4$

7. Larger: $3/5$ or $5/8$ — type it

8. Is $5/9$ more or less than $1/2$

9. $3/4$ as a decimal

10. Larger: $2/3$ or $5/8$ — type it

11. Smallest common denominator of $1/4$ and $1/6$



Working space



Error journal

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Common Denominators

OUT OF 21

Warm-Up • 1-11 • 4 | 6 | 6 | 2 | 2 | 8 | 6 | 9 | 3 | 3 | 2

Core • 12-21 • 12 | 10 | 8 | 9 | 24 | 12 | 15 | 10 | 24 | 24

Challenge • 22-25 • 24 | 30 | 12 | 36

3.2 • Benchmarks

OUT OF 19

Warm-Up • 1-12 • 0 | 1 | $\frac{1}{2}$ | 1 | 0 | more | 0 | 1 | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | $\frac{1}{2}$

Core • 13-19 • more | less | less | more | 1 | more | 1

Challenge • 20-23 • more | $\frac{1}{2}$ | less | $\frac{1}{2}$

3.3 • Comparing & Ordering

OUT OF 22

Warm-Up • 1-12 • $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{1}{4}$ | $\frac{2}{3}$ | $\frac{1}{6}$ | $\frac{5}{8}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{3}{5}$ | $\frac{1}{3}$ | $\frac{1}{10}$ | $\frac{5}{6}$

Core • 13-22 • $\frac{3}{4}$ | $\frac{5}{8}$ | $\frac{5}{6}$ | $\frac{1}{4}$ | $\frac{2}{3}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{7}{12}$ | $\frac{4}{5}$ | $\frac{3}{8}$

Challenge • 23-26 • $\frac{7}{8}$ | $\frac{11}{15}$ | $\frac{7}{10}$ | $\frac{2}{3}$

3.4 • On the Number Line

OUT OF 17

Warm-Up • 1-10 • $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | 3 | 4 | 0.5 | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{7}{8}$ | 4

Core • 11-17 • 0.75 | 0.25 | 0.2 | 0.6 | $\frac{3}{8}$ | 0.125 | 0.1

Challenge • 18-21 • 0.875 | $\frac{1}{2}$ | 0.375 | $\frac{1}{2}$

Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 4 | $\frac{1}{2}$ | 3 | $\frac{3}{4}$

Core • 5-11 • $\frac{3}{4}$ | 12 | $\frac{5}{8}$ | more | 0.75 | $\frac{2}{3}$ | 12

Challenge • 12-15 • $\frac{7}{8}$ | 0.375 | less | $\frac{3}{4}$

Adding Like Denominators

Same size pieces already. Add the counts, leave the bottom alone.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{4} + \frac{1}{4}$ — type as a/b

2 $\frac{1}{5} + \frac{2}{5}$

3 $\frac{1}{8} + \frac{3}{8}$

4 $\frac{2}{6} + \frac{1}{6}$

5 $\frac{1}{3} + \frac{1}{3}$

6 $\frac{3}{10} + \frac{2}{10}$

7 $\frac{1}{6} + \frac{1}{6}$ — type as a/b

8 $\frac{2}{7} + \frac{3}{7}$

9 $\frac{1}{10} + \frac{4}{10}$

10 $\frac{3}{9} + \frac{3}{9}$

11 $\frac{1}{12} + \frac{5}{12}$

12 $\frac{2}{8} + \frac{2}{8}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{3}{8} + \frac{3}{8}$

14. $\frac{5}{12} + \frac{3}{12}$

15. $\frac{7}{10} + \frac{1}{10}$

16. $\frac{2}{9} + \frac{4}{9}$

17. $\frac{5}{6} + \frac{1}{6}$

18. $\frac{4}{15} + \frac{6}{15}$

19. $\frac{7}{20} + \frac{3}{20}$

20. $\frac{5}{9} + \frac{1}{9}$

21. $\frac{3}{14} + \frac{4}{14}$



Subtracting Like Denominators

Same move, other direction.

Warm-Up 10 ITEMS • RECALL

1 $3/4 - 1/4$ — type as a/b

2 $4/5 - 1/5$

3 $7/8 - 3/8$

4 $5/6 - 1/6$

5 $2/3 - 1/3$

6 $9/10 - 4/10$

7 $5/8 - 1/8$ — type as a/b

8 $5/7 - 2/7$

9 $4/6 - 1/6$

10 $8/10 - 3/10$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

11. $1 - 1/4$

12. $1 - 3/8$

13. $11/12 - 5/12$

14. $1 - 2/5$

15. $7/9 - 1/9$

16. $7/12 - 1/12$

17. $11/15 - 1/15$

18. $13/20 - 3/20$

19. $5/6 - 1/2$

20. $7/8 - 1/4$



Unlike Denominators

The grade-five ceiling. Convert first, then add.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2} + \frac{1}{4}$ — type as a/b

2 $\frac{1}{3} + \frac{1}{6}$

3 $\frac{1}{2} + \frac{1}{8}$

4 $\frac{1}{4} + \frac{1}{8}$

5 $\frac{1}{2} - \frac{1}{4}$

6 $\frac{2}{3} - \frac{1}{6}$

7 $\frac{1}{2} + \frac{1}{6}$ — type as a/b

8 $\frac{1}{4} + \frac{1}{2}$

9 $\frac{1}{3} + \frac{1}{9}$

10 $\frac{1}{5} + \frac{1}{10}$

11 $\frac{3}{4} - \frac{1}{8}$

12 $\frac{1}{2} - \frac{1}{6}$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $\frac{1}{2} + \frac{1}{3}$

14. $\frac{3}{4} + \frac{1}{8}$

15. $\frac{2}{3} + \frac{1}{4}$

16. $\frac{3}{4} - \frac{1}{3}$

17. $\frac{5}{6} - \frac{1}{2}$

18. $\frac{2}{3} + \frac{1}{6}$

19. $\frac{3}{5} + \frac{1}{10}$

20. $\frac{5}{8} - \frac{1}{4}$

21. $\frac{3}{4} - \frac{2}{3}$

22. $\frac{2}{5} + \frac{1}{4}$



Working space



Error journal

Fractions in Words

Read it twice, then decide whether it is an add or a subtract.

Warm-Up 12 ITEMS • RECALL

1 Ate $\frac{1}{4}$ of a pizza, then $\frac{1}{4}$ more — total, as a/b

2 Had $\frac{3}{4}$, ate $\frac{1}{4}$ — left

3 $\frac{1}{2}$ hour plus $\frac{1}{2}$ hour — hours

4 $\frac{2}{5}$ of a jug plus $\frac{1}{5}$ — as a/b

5 1 whole minus $\frac{1}{3}$ — as a/b

6 $\frac{1}{8} + \frac{1}{8}$ — as a/b

7 Drank $\frac{1}{3}$ of a bottle, then $\frac{1}{3}$ more — total, as a/b

8 Had $\frac{5}{6}$, used $\frac{1}{6}$ — left

9 $\frac{1}{4}$ hour plus $\frac{3}{4}$ hour — hours

10 $\frac{3}{8}$ of a tank plus $\frac{1}{8}$ — as a/b

11 Read $\frac{1}{2}$ a book, then $\frac{1}{4}$ — as a/b

12 Had $\frac{7}{8}$, gave away $\frac{3}{8}$ — as a/b

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Walked $\frac{1}{2}$ mile then $\frac{1}{4}$ mile — total, as a/b

14. A $\frac{3}{4}$ cup recipe, you have $\frac{1}{2}$ — how much more, as a/b

15. Read $\frac{2}{3}$ of a book, then $\frac{1}{6}$ more — total, as a/b

16. Ran $\frac{5}{6}$ km and walked $\frac{1}{3}$ km — total, as a/b

17. 1 hour minus $\frac{1}{4}$ hour — minutes

18. Walked $\frac{2}{5}$ km then $\frac{1}{2}$ km — total as a/b

19. $\frac{3}{4}$ of an hour in minutes

20. $\frac{1}{3}$ of an hour in minutes

21. Ate $\frac{3}{8}$ of a pie, dad ate $\frac{1}{4}$ — total as a/b

22. $1 - \frac{5}{8}$ as a/b



Working space



Error journal

Closest to One

Draw cards, build a fraction, get nearest to 1 without going over.

✦ Warm-Up 4 ITEMS • RECALL

1 Which is nearer 1: $\frac{3}{4}$ or $\frac{1}{2}$ — type it

2 Which is nearer 1: $\frac{7}{8}$ or $\frac{3}{4}$

3 Which is nearer 1: $\frac{5}{6}$ or $\frac{2}{3}$ — type it

4 Which is nearer 1: $\frac{9}{10}$ or $\frac{4}{5}$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. From digits 5 and 6, the fraction under 1 — type as a/b

.....

6. From 7 and 8, the fraction under 1

.....

7. Which is nearer 1: $\frac{7}{8}$ or $\frac{8}{9}$ — type it

.....

8. Which is nearer 1: $\frac{11}{12}$ or $\frac{5}{6}$

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Adding Like Denominators

OUT OF 21

Warm-Up • 1-12 • $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{5}{7}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{1}{2}$

Core • 13-21 • $\frac{3}{4}$ | $\frac{2}{3}$ | $\frac{4}{5}$ | $\frac{2}{3}$ | 1 | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{2}$

Challenge • 22-25 • $\frac{3}{2}$ | $\frac{7}{5}$ | $\frac{5}{3}$ | $\frac{8}{5}$

4.2 • Subtracting Like Denominators

OUT OF 20

Warm-Up • 1-10 • $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{3}$ | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{3}{7}$ | $\frac{1}{2}$ | $\frac{1}{2}$

Core • 11-20 • $\frac{3}{4}$ | $\frac{5}{8}$ | $\frac{1}{2}$ | $\frac{3}{5}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | $\frac{5}{8}$

Challenge • 21-24 • $\frac{5}{4}$ | $\frac{3}{4}$ | $\frac{7}{4}$ | $\frac{3}{2}$

4.3 • Unlike Denominators

OUT OF 22

Warm-Up • 1-12 • $\frac{3}{4}$ | $\frac{1}{2}$ | $\frac{5}{8}$ | $\frac{3}{8}$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{4}{9}$ | $\frac{3}{10}$ | $\frac{5}{8}$ | $\frac{1}{3}$

Core • 13-22 • $\frac{5}{6}$ | $\frac{7}{8}$ | $\frac{11}{12}$ | $\frac{5}{12}$ | $\frac{1}{3}$ | $\frac{5}{6}$ | $\frac{7}{10}$ | $\frac{3}{8}$ | $\frac{1}{12}$ | $\frac{13}{20}$

Challenge • 23-26 • $\frac{17}{12}$ | 1 | $\frac{19}{12}$ | $\frac{7}{8}$

4.4 • Fractions in Words

OUT OF 22

Warm-Up • 1-12 • $\frac{1}{2}$ | $\frac{1}{2}$ | 1 | $\frac{3}{5}$ | $\frac{2}{3}$ | $\frac{1}{4}$ | $\frac{2}{3}$ | $\frac{2}{3}$ | 1 | $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{1}{2}$

Core • 13-22 • $\frac{3}{4}$ | $\frac{1}{4}$ | $\frac{5}{6}$ | $\frac{7}{6}$ | 45 | $\frac{9}{10}$ | 45 | 20 | $\frac{5}{8}$ | $\frac{3}{8}$

Challenge • 23-26 • $\frac{9}{8}$ | 40 | $\frac{8}{5}$ | 50

Fri • Closest to One

OUT OF 8

Warm-Up • 1-4 • $\frac{3}{4}$ | $\frac{7}{8}$ | $\frac{5}{6}$ | $\frac{9}{10}$

Core • 5-8 • $\frac{5}{6}$ | $\frac{7}{8}$ | $\frac{8}{9}$ | $\frac{11}{12}$

Challenge • 9-14 • $\frac{9}{10}$ | $\frac{1}{10}$ | $\frac{5}{6}$ | $\frac{7}{8}$ | $\frac{4}{5}$ | $\frac{1}{9}$

Fraction of a Set

Divide by the bottom, multiply by the top.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2}$ of 10
_____2 $\frac{1}{3}$ of 12
_____3 $\frac{1}{4}$ of 20
_____4 $\frac{1}{5}$ of 25
_____5 $\frac{1}{2}$ of 30
_____6 $\frac{1}{6}$ of 30
_____7 $\frac{1}{2}$ of 14
_____8 $\frac{1}{3}$ of 15
_____9 $\frac{1}{4}$ of 24
_____10 $\frac{1}{5}$ of 35
_____11 $\frac{1}{6}$ of 36
_____12 $\frac{1}{8}$ of 40

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

13. $\frac{2}{5}$ of 30
.....14. $\frac{3}{4}$ of 20
.....15. $\frac{2}{3}$ of 18
.....16. $\frac{5}{6}$ of 24
.....17. $\frac{3}{8}$ of 40
.....18. $\frac{3}{4}$ of 32
.....

Of Money and Time

The same operator, applied to dollars and minutes.

✦ Warm-Up 9 ITEMS • RECALL

1 $\frac{1}{2}$ of \$20

2 $\frac{1}{4}$ of \$40

3 $\frac{1}{2}$ of an hour in minutes

4 $\frac{1}{4}$ of an hour in minutes

5 $\frac{1}{3}$ of \$30

6 $\frac{1}{5}$ of \$50

7 $\frac{1}{2}$ of \$10

8 $\frac{1}{4}$ of \$20

9 $\frac{1}{5}$ of \$25

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. $\frac{3}{4}$ of \$60

.....

11. $\frac{2}{3}$ of an hour in minutes

.....

12. $\frac{3}{5}$ of \$45

.....

13. $\frac{5}{6}$ of an hour in minutes

.....

14. $\frac{2}{5}$ of \$75

.....

15. $\frac{2}{3}$ of \$45

.....

16. $\frac{3}{4}$ of an hour in minutes

.....

17. $\frac{2}{5}$ of \$50

.....

★ Challenge NOT GRADED • REASON IT OUT18. $\frac{3}{8}$ of \$9619. $\frac{7}{12}$ of an hour in minutes**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Working Backwards

Given the part, find the whole.

Warm-Up 12 ITEMS • RECALL

1 Half a number is 6. The number

2 A third is 4. The number

3 A quarter is 5. The number

4 A fifth is 3. The number

5 Half is 15. The number

6 A sixth is 2. The number

7 $\frac{1}{2}$ of 100

8 $\frac{1}{4}$ of 100

9 $\frac{1}{10}$ of 100

10 $\frac{1}{5}$ of 100

11 $\frac{3}{4}$ of 100

12 $\frac{1}{2}$ of 60

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $\frac{2}{3}$ of a number is 12. The number
.....

14. $\frac{3}{4}$ of a number is 15. The number
.....

15. $\frac{2}{5}$ of a number is 10. The number
.....

16. $\frac{5}{6}$ of a number is 20. The number
.....

17. $\frac{3}{8}$ of a number is 9. The number
.....

18. $\frac{2}{5}$ of 200
.....

19. $\frac{3}{10}$ of 150
.....

20. $\frac{5}{8}$ of 160
.....

21. $\frac{2}{3}$ of 150
.....

22. $\frac{4}{5}$ of 250
.....



Working space



Error journal

Mixed Fraction Problems

No signposting. Read, decide, then compute.

Warm-Up 12 ITEMS • RECALL

1 $\frac{1}{2}$ of 16
_____2 $\frac{1}{4} + \frac{1}{4}$ — type as a/b
_____3 $1 - \frac{1}{2}$ — as a/b
_____4 $\frac{2}{6}$ simplified — as a/b
_____5 $\frac{1}{3}$ of 15
_____6 $\frac{3}{4} - \frac{1}{4}$ — as a/b
_____7 $\frac{1}{2}$ of 8
_____8 $\frac{1}{3}$ of 9
_____9 $\frac{1}{4}$ of 16
_____10 $\frac{1}{2}$ of 22
_____11 $\frac{1}{5}$ of 20
_____12 $\frac{1}{6}$ of 24

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $\frac{2}{3} + \frac{1}{6}$ — as a/b
.....14. $\frac{3}{4}$ of 24
.....15. $\frac{5}{8} - \frac{1}{4}$ — as a/b
.....16. $\frac{2}{5}$ of a number is 8. The number
.....17. Larger: $\frac{5}{8}$ or $\frac{2}{3}$ — type it
.....18. $\frac{3}{4}$ of 28
.....19. $\frac{2}{3}$ of 33
.....20. $\frac{4}{5}$ of 45
.....21. $\frac{5}{8}$ of 48
.....22. $\frac{3}{7}$ of 42
.....



Working space



Error journal

Fold the Universe, Built

Finish the paper number line with every fraction landed on it.

Warm-Up 4 ITEMS • RECALL

1 Folds to get halves

2 Folds to get quarters

3 $\frac{1}{2}$ of 12

4 $\frac{1}{4}$ of 12

Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Folds to get eighths

.....

6. Sections after 4 folds

.....

7. $\frac{2}{3}$ of 24

.....

8. $\frac{3}{5}$ of 40

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Fraction of a Set

OUT OF 18

Warm-Up • 1-12 • 5 | 4 | 5 | 5 | 15 | 5 | 7 | 5 | 6 | 7 | 6 | 5

Core • 13-18 • 12 | 15 | 12 | 20 | 15 | 24

Challenge • 19-22 • 30 | 84 | 51 | 105

5.2 • Of Money and Time

OUT OF 17

Warm-Up • 1-9 • 10 | 10 | 30 | 15 | 10 | 10 | 5 | 5 | 5

Core • 10-17 • 45 | 40 | 27 | 50 | 30 | 30 | 45 | 20

Challenge • 18-19 • 36 | 35

5.3 • Working Backwards

OUT OF 22

Warm-Up • 1-12 • 12 | 12 | 20 | 15 | 30 | 12 | 50 | 25 | 10 | 20 | 75 | 30

Core • 13-22 • 18 | 20 | 25 | 24 | 24 | 80 | 45 | 100 | 100 | 200

Challenge • 23-26 • 63 | 50 | 140 | 60

5.4 • Mixed Fraction Problems

OUT OF 22

Warm-Up • 1-12 • 8 | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{3}$ | 5 | $\frac{1}{2}$ | 4 | 3 | 4 | 11 | 4 | 4

Core • 13-22 • $\frac{5}{6}$ | 18 | $\frac{3}{8}$ | 20 | $\frac{2}{3}$ | 21 | 22 | 36 | 30 | 18

Challenge • 23-26 • $\frac{17}{12}$ | 35 | 225 | 60

Fri • Fold the Universe, Built

OUT OF 8

Warm-Up • 1-4 • 1 | 2 | 6 | 3

Core • 5-8 • 3 | 16 | 16 | 24

Challenge • 9-14 • 64 | 5 | 15 | 84 | 110 | 183

The Equivalence Proof

Show the same amount two different ways and say why they match.

Warm-Up 12 ITEMS • RECALL

1 $1/2 = ?/4$ — the numerator

2 $2/4$ simplified — type as a/b

3 $1/2 \times 3/3$ — type as a/b

4 $3/6$ simplified

5 $3/3$ as a whole number

6 Multiplying by 1 changes a number by how much

7 $1/2 + 1/2$ — type as a/b

8 $1/4 + 1/4$

9 $2/3 - 1/3$

10 $1/2$ of 16

11 $3/4$ of 8

12 $5/8 - 1/8$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $2/3 \times 4/4$ — type as a/b

.....

14. $8/12$ simplified

.....

15. $5/6 \times 3/3$

.....

16. $15/18$ simplified

.....

17. What $3/4$ was multiplied by to become $12/16$

.....

18. $1/2 + 1/3$

.....

19. $3/4 - 1/3$

.....

20. $2/3$ of 27

.....

21. $5/6 + 1/3$

.....

22. $7/10 - 1/5$

.....



Working space



Error journal

Number Line Defence

Every fraction has one place. Put it there and argue for it.

Warm-Up 12 ITEMS • RECALL

1 $1/2$ as a decimal

2 $1/4$ as a decimal

3 $3/4$ as a decimal

4 $1/5$ as a decimal

5 $2/5$ as a decimal

6 $1/10$ as a decimal

7 $1/3$ of 12

8 $1/2$ of 18

9 $1/5$ of 30

10 $1/4 + 1/2$

11 $3/5 - 1/5$

12 $1/6$ of 42

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $3/8$ as a decimal

.....

14. $5/8$ as a decimal

.....

15. $7/10$ as a decimal

.....

16. Which sits further right: $2/3$ or 0.6 — type it

.....

17. $3/5$ as a decimal

.....

18. $2/5 + 1/2$

.....

19. $5/6 - 1/4$

.....

20. $3/8$ of 88

.....

21. $4/9 + 1/3$

.....

22. $7/12 - 1/4$

.....



Working space



Error journal

Mission Review

Everything from six weeks, mixed together.

Warm-Up 12 ITEMS • RECALL

1 $6/8$ simplified — type as a/b

2 $1/4 + 1/4$ — as a/b

3 $1/2$ of 14

4 $1 - 1/3$ — as a/b

5 Larger: $1/2$ or $1/3$ — type it

6 $1/2$ as a decimal

7 Half of a half — type as a/b

8 Half of a quarter

9 Two quarters

10 Three thirds

11 $1/2$ of 40

12 $1/4$ of 40

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $2/3 + 1/4$ — as a/b

14. $3/4$ of 32

15. Common denominator of $5/6$ and $3/8$

16. $3/4 - 1/3$ — as a/b

17. $2/5$ of a number is 10. The number

18. $1/2$ of $1/3$ — type as a/b

19. $1/3$ of $1/4$

20. $2/3$ of $1/2$

21. $3/4$ of $1/2$

22. $1/5$ of $1/2$



Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 $2/4$ simplified — type as a/b

2 $1/3 + 1/3$ — as a/b

3 $1/4$ of 16

4 Is $3/6$ the same as $1/2$ — yes or no

5 $1/5$ as a decimal

6 $5/6 - 1/6$ — as a/b

7 $1/2 + 1/4$

8 $2/3$ of 9

9 $5/6 - 1/6$

10 $1/4$ of 32

11 $3/8 + 1/8$

12 $1/3$ of 24

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $1/2 + 1/3$ — as a/b

14. $12/16$ simplified

15. $5/6$ of 24

16. Larger: $2/3$ or $3/4$ — type it

17. $3/8$ as a decimal

18. $3/4 + 1/6$

19. $7/8 - 1/2$

20. $5/8$ of 64

21. $2/5 + 3/10$

22. $2/3$ of 42



Working space



Error journal

Mission 04 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 $\frac{4}{8}$ simplified — type as a/b
_____2 $\frac{1}{4} + \frac{1}{4}$ — as a/b
_____3 $\frac{1}{2}$ of 20
_____4 $\frac{1}{4} + \frac{1}{4}$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.5. $\frac{1}{2} = \frac{?}{10}$ — the numerator
.....6. $\frac{10}{15}$ simplified
.....7. $\frac{1}{2} + \frac{1}{4}$ — as a/b
.....8. $\frac{3}{4} - \frac{1}{3}$ — as a/b
.....9. Larger: $\frac{3}{5}$ or $\frac{5}{8}$ — type it
.....10. $\frac{2}{3}$ of 18
.....11. $\frac{3}{4}$ as a decimal
.....12. Common denominator of $\frac{2}{3}$ and $\frac{3}{4}$
.....13. $\frac{2}{3} + \frac{1}{4}$
.....14. $\frac{3}{4}$ of 44
.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 6 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

6.1 • The Equivalence Proof

OUT OF 22

Warm-Up • 1-12 • 2 | $\frac{1}{2}$ | $\frac{3}{6}$ | $\frac{1}{2}$ | 1 | 0 | 1 | $\frac{1}{2}$ | $\frac{1}{3}$ | 8 | 6 | $\frac{1}{2}$

Core • 13-22 • $\frac{8}{12}$ | $\frac{2}{3}$ | $\frac{15}{18}$ | $\frac{5}{6}$ | 4 | $\frac{5}{6}$ | $\frac{5}{12}$ | 18 | $\frac{7}{6}$ | $\frac{1}{2}$

Challenge • 23-26 • $\frac{2}{3}$ | $\frac{2}{3}$ | $\frac{17}{12}$ | 95

6.2 • Number Line Defence

OUT OF 22

Warm-Up • 1-12 • 0.5 | 0.25 | 0.75 | 0.2 | 0.4 | 0.1 | 4 | 9 | 6 | $\frac{3}{4}$ | $\frac{2}{5}$ | 7

Core • 13-22 • 0.375 | 0.625 | 0.7 | $\frac{2}{3}$ | 0.6 | $\frac{9}{10}$ | $\frac{7}{12}$ | 33 | $\frac{7}{9}$ | $\frac{1}{3}$

Challenge • 23-26 • 0.875 | $\frac{1}{3}$ | $\frac{41}{24}$ | 132

6.3 • Mission Review

OUT OF 22

Warm-Up • 1-12 • $\frac{3}{4}$ | $\frac{1}{2}$ | 7 | $\frac{2}{3}$ | $\frac{1}{2}$ | 0.5 | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{2}$ | 1 | 20 | 10

Core • 13-22 • $\frac{11}{12}$ | 24 | 24 | $\frac{5}{12}$ | 25 | $\frac{1}{6}$ | $\frac{1}{12}$ | $\frac{1}{3}$ | $\frac{3}{8}$ | $\frac{1}{10}$

Challenge • 23-26 • 1 | 35 | $\frac{1}{2}$ | $\frac{1}{2}$

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • $\frac{1}{2}$ | $\frac{2}{3}$ | 4 | yes | 0.2 | $\frac{2}{3}$ | $\frac{3}{4}$ | 6 | $\frac{2}{3}$ | 8 | $\frac{1}{2}$ | 8

Core • 13-22 • $\frac{5}{6}$ | $\frac{3}{4}$ | 20 | $\frac{3}{4}$ | 0.375 | $\frac{11}{12}$ | $\frac{3}{8}$ | 40 | $\frac{7}{10}$ | 28

Challenge • 23-26 • $\frac{17}{12}$ | 50 | $\frac{29}{24}$ | 84

Fri • Mission 04 Test

OUT OF 14

Warm-Up • 1-4 • $\frac{1}{2}$ | $\frac{1}{2}$ | 10 | $\frac{1}{2}$

Core • 5-14 • 5 | $\frac{2}{3}$ | $\frac{3}{4}$ | $\frac{5}{12}$ | $\frac{5}{8}$ | 12 | 0.75 | 12 | $\frac{11}{12}$ | 33

Challenge • 15-19 • $\frac{17}{12}$ | 24 | $\frac{17}{12}$ | 154 | $\frac{2}{5}$